

ANALYSIS REPORT

August 17nd 2021

Your reference: Various samples of Dirk Paessler received on 02/07/2021

Our reference: 2107AD

Table 1: Sample list with requested analyses.

Our reference	Your reference	XRF	XRD	PSD	CEC	TIC/ TOC	BET-N2
2107AD01	Fürth 1					X	
2107AD02	Fürth 2					X	
2107AD03	Fürth 3					X	
2107AD04	Fürth 4					X	
2107AD05	Fürth 5					X	
2107AD06	N Downhill	X	X	X	X		X
2107AD07	S Uphill	X	X	X	X		X

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Table 2: Measurement characteristics of the BET analysis.

Parameter	Value
Instrument	Quantachrome Autosorb
Date of analysis	09/07/2021 – 10/07/2021
Date of data treatment	02/08/2021
Operator	A.A.
Sample preparation	Outgassing for at least 24h at 200°C under high-vacuum
Adsorptive-gas	N ₂
Temperature during analysis	77.35K (liquid nitrogen)
Results	Table 8 and Figure 1

Table 3: Measurement characteristics of the CEC measurement.

Parameter	Value
Analysis	Cation Exchange Capacity
General sample handling	Drying at 40°C
Sample preparation	Grinding <500µm, exchange with Co-Hexamine Trichloride
Instrument	Shimadzu UV-1280
Measured by	A.A.
Date of measurement	20/07/2021
Results	Table 9

Table 4: Measurement characteristics of the TOC/TIC analysis.

Parameter	Value
Analysis	Total Organic Carbon and Inorganic Carbon analysis
General sample handling	Each sample was splitted in two representative parts: one for TOC analysis, one for TIC analysis. The parts for TOC analysis were sequentially treated with HCl to remove the inorganic carbon fraction.
Sample preparation	The two parts of each sample were fused at 1600-1800°C and measured by chromatography
Apparatus	Carlo Erba EA1108 Elemental Analyzer
Date of measurement	15/07/2021
Results	Table 10

Table 5: Measurement characteristics of the bulk XRD analysis.

Parameter	Value
Analysis	Bulk mineralogical analysis by X-ray diffraction
General sample handling	Drying at 40°C, homogenisation by mortar & pestle
Sample preparation	Wet milling and drying
Diffractionmeter	Bruker D8 Advance, XE-T detector, Cu-Kα radiation
Data treatment methodology	In-house
Interpretation by	Dr. Gilles Mertens
Date of measurement	09/07/2021
Date of data treatment	02/08/2021
Results	Table 11 and Figures 2 and 3

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Table 6: Measurement characteristics of the XRF analysis.

Parameter	Value
Analysis	X-ray Fluorescence analysis
Sample preparation	Prior to fusion, the loss on ignition (LOI), which includes H ₂ O, CO ₂ , S and other volatiles, is determined from the weight loss after roasting the sample at 1000°C for 2 hours. The fusion disk is made by mixing a 0.75 g equivalent of the roasted sample with 9.75 g of a combination of lithium metaborate and lithium tetraborate with lithium bromide as a releasing agent. The samples were fused in a Pt crucible using an automated crucible fluxer and automatically poured into a Pt mold for casting.
Apparatus	Panalytical Axios Advanced wavelength dispersive XRF
Date of measurement	14/07/2021
Results	Table 11

Table 7: Characteristics of the grain-size analysis.

Parameter	Value
Analysis	Laser diffraction
Sample preparation	Dispersion in water and splitting the sample – individual splits are added to the instrument until a proper obscuration is obtained
Apparatus	Fritsch Analysette A-22 NeXt
Use of ultrasonic dispersion	Ja
Model	Mie theory
Measured by	C.D.
Date of analysis	12/08/2021
Results	Table 12 & 13 and Figure 4

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Table 8: BET specific surface area: results.

Sample	Sample mass (g)	Correlation coefficient of BET plot	BET area (m ² /g)
N Downhill	2.1042	0.999974	5.0
S Uphill	1.7366	0.999940	5.6

Table 9: Cation exchange capacity: results.

Sample	CEC (meq/100g)
N Downhill	6.0
S Uphill	5.2

Table 10: Results of the total organic carbon analysis (TOC) and total inorganic carbon analysis (TIC). Results are expressed in weight percentages.

Analysis	Unit	Fürth 1	Fürth 2	Fürth 3	Fürth 4	Fürth 5
TIC	%	0.05	0.06	0.00	0.00	0.09
TOC	%	1.30	1.37	1.30	1.73	1.24

Table 11: Quantitative bulk mineralogical compositions of the samples (in weight percentages of the identified crystalline components).

Mineral	Theoretical formula ¹	N Downhill	S Uphill
Silicates			
Quartz	SiO ₂	63.3	57.8
K-feldspar	(K,Na)(Si,Al) ₄ O ₈	24.1	26.6
Plagioclase/Albite	(Ca,Na)(Si,Al) ₄ O ₈	3.7	3.9
Oxides			
Anatase	TiO ₂	0.2	0.2
Hematite	Fe ₂ O ₃	0.3	0.4
Layer silicates			
2:1 layer silicates	(Na,Ca,K) _{0,3} (Mg,Fe,Al) ₃ Si ₄ O ₁₀ (OH) ₂ .nH ₂ O	7.5	10.0
Chlorite	(Mg,Fe) ₅ Al(Si ₃ Al)O ₁₀ (OH) ₈	0.8	1.1

¹ These formulae are general formulae and do not necessarily correspond to the composition of the minerals in these specific samples.

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Table 11: Results of the chemical XRF analysis of sample "0-30cm | 03/05/21". The Loss on Ignition (LOI) was determined on the fused samples.

Element	Unit	Detection limit	N Downhill	S Uphill
SiO ₂	%	0.01	81.81	81.21
Al ₂ O ₃	%	0.01	7.29	8.07
Fe ₂ O ₃	%	0.01	1.12	1.26
MnO	%	0.01	0.105	0.091
MgO	%	0.01	0.33	0.41
CaO	%	0.01	0.26	0.25
Na ₂ O	%	0.01	0.47	0.51
K ₂ O	%	0.01	4.08	4.44
TiO ₂	%	0.01	0.22	0.23
P ₂ O ₅	%	0.01	0.19	0.19
Cr ₂ O ₃	%	0.01	0.01	< 0.01
V ₂ O ₅	%	0.003	0.004	0.005
Co ₃ O ₄	%	0.005	< 0.005	< 0.005
CuO	%	0.005	< 0.005	< 0.005
NiO	%	0.003	< 0.003	< 0.003
LOI	%		3.36	3.26
TOTAL	%		99.24	99.93

Table 12: Grain-size analysis by laser diffraction of the samples.

Size (µm)	N Downhill	S Uphill
0.01	0.0	0.0
0.01	0.0	0.0
0.01	0.0	0.0
0.01	0.0	0.0
0.02	0.0	0.0
0.02	0.0	0.0
0.02	0.0	0.0
0.02	0.0	0.0
0.02	0.0	0.0
0.03	0.0	0.0
0.03	0.0	0.0
0.03	0.0	0.0
0.04	0.0	0.0
0.04	0.0	0.0
0.05	0.0	0.0
0.05	0.0	0.0
0.06	0.0	0.0
0.07	0.0	0.0
0.07	0.0	0.0
0.08	0.0	0.0
0.09	0.0	0.0
0.10	0.0	0.0
0.11	0.0	0.0
0.12	0.0	0.0

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0.14	0.0	0.0
0.15	0.0	0.0
0.17	0.0	0.0
0.19	0.0	0.0
0.21	0.0	0.0
0.24	0.0	0.0
0.26	0.0	0.0
0.29	0.0	0.0
0.32	0.0	0.0
0.36	0.0	0.0
0.40	0.0	0.0
0.45	0.0	0.0
0.50	0.0	0.0
0.56	0.0	0.0
0.62	0.0	0.0
0.69	0.0	0.0
0.77	0.1	0.0
0.85	0.1	0.1
0.95	0.3	0.2
1.06	0.5	0.4
1.17	0.9	0.6
1.31	1.4	1.0
1.45	2.0	1.5
1.62	2.9	2.1
1.80	3.9	2.8
2.01	5.1	3.6
2.23	6.4	4.6
2.48	7.9	5.6
2.77	9.6	6.8
3.08	11.3	8.0
3.43	13.1	9.3
3.81	15.0	10.7
4.24	16.8	12.0
4.72	18.7	13.4
5.26	20.5	14.8
5.85	22.2	16.2
6.51	23.8	17.6
7.25	25.3	18.9
8.07	26.8	20.2
8.98	28.2	21.5
9.99	29.5	22.9
11.12	30.9	24.2
12.38	32.2	25.5
13.78	33.6	26.9
15.34	35.0	28.2
17.07	36.5	29.5
19.00	38.1	30.7
21.15	39.6	31.9
23.54	41.2	32.9
26.20	42.7	33.8
29.16	44.2	34.6
32.45	45.6	35.2
36.12	46.8	35.7

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40.20	47.9	36.2
44.74	48.8	36.6
49.80	49.6	37.0
55.43	50.2	37.5
61.69	50.7	38.0
68.66	51.1	38.7
76.42	51.4	39.4
85.06	51.6	40.3
94.67	51.9	41.2
105.37	52.2	42.2
117.28	52.7	43.3
130.53	53.6	44.5
145.28	54.9	46.1
161.70	56.8	48.0
179.97	59.5	50.6
200.31	62.9	53.8
222.95	67.0	57.8
248.14	71.7	62.5
276.18	76.9	67.9
307.40	82.1	73.7
342.13	87.1	79.5
380.80	91.5	85.1
423.83	95.0	90.0
471.73	97.4	93.9
525.04	98.9	96.8
584.37	99.7	98.6
650.41	99.9	99.5
723.91	100.0	99.9
805.72	100.0	100.0
896.77	100.0	100.0
998.11	100.0	100.0
1110.90	100.0	100.0
1236.44	100.0	100.0
1376.17	100.0	100.0
1531.69	100.0	100.0
1704.78	100.0	100.0
1897.44	100.0	100.0
2111.86	100.0	100.0

Table 13: Statistical parameters of the laser diffraction measurements.

Parameter	N Downhill	S Uphill
Mean (µm)	152.12	196.41
Standard Deviation (µm)	162.47	181.1
Skewness	0.73	0.55
Kurtosis	-0.63	-0.71
d10 (µm)	3.00	3.82
d50 (µm)	56.29	185.16
d90 (µm)	387.43	446.63

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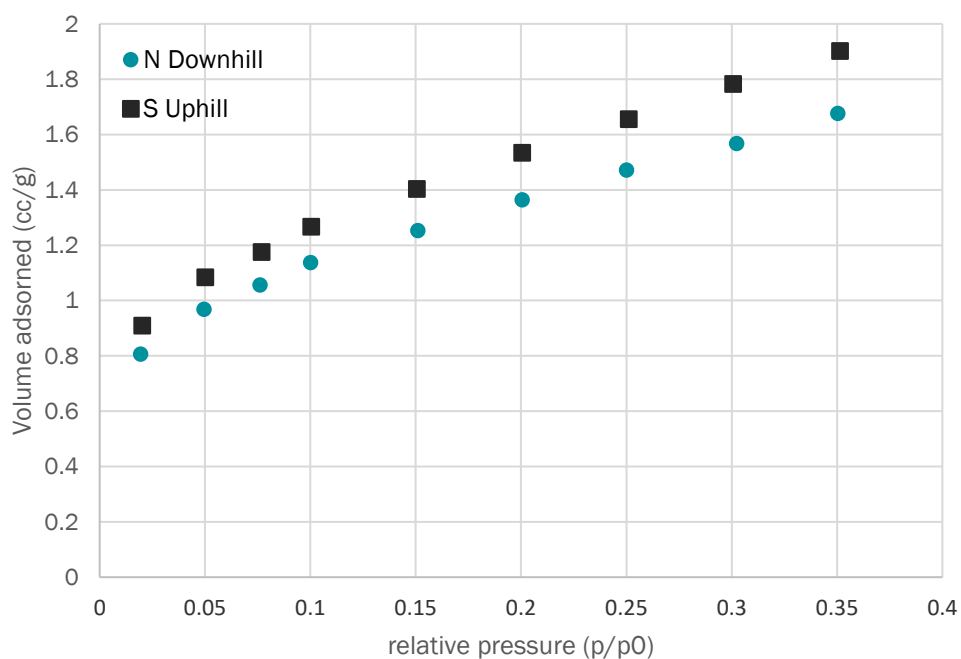


Figure 1: Isotherms of the BET analysis of the samples.

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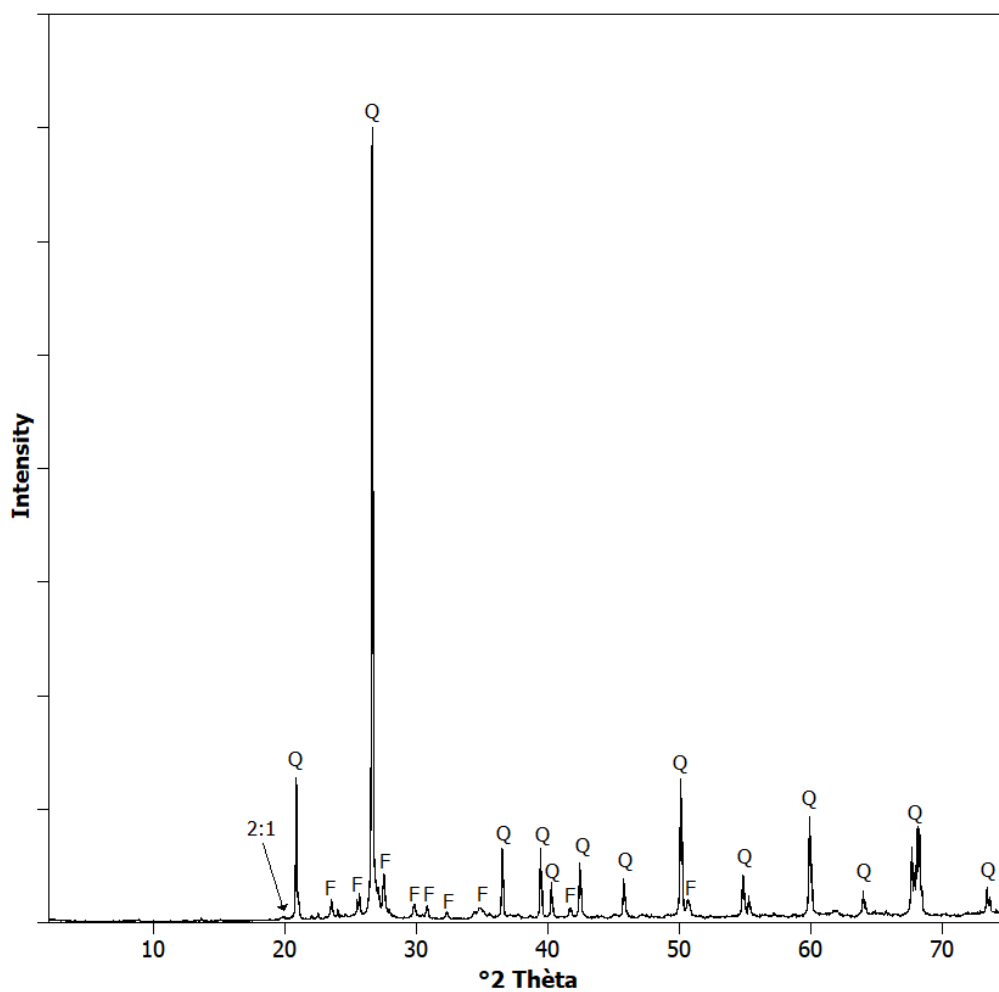


Figure 2: Diffraction pattern of the sample “N Downhill”. The minerals that contribute to the main reflections are labeled: 2:1 layer silicates (2:1), Quartz (Q), Alkali feldspar and Plagioclase (F).

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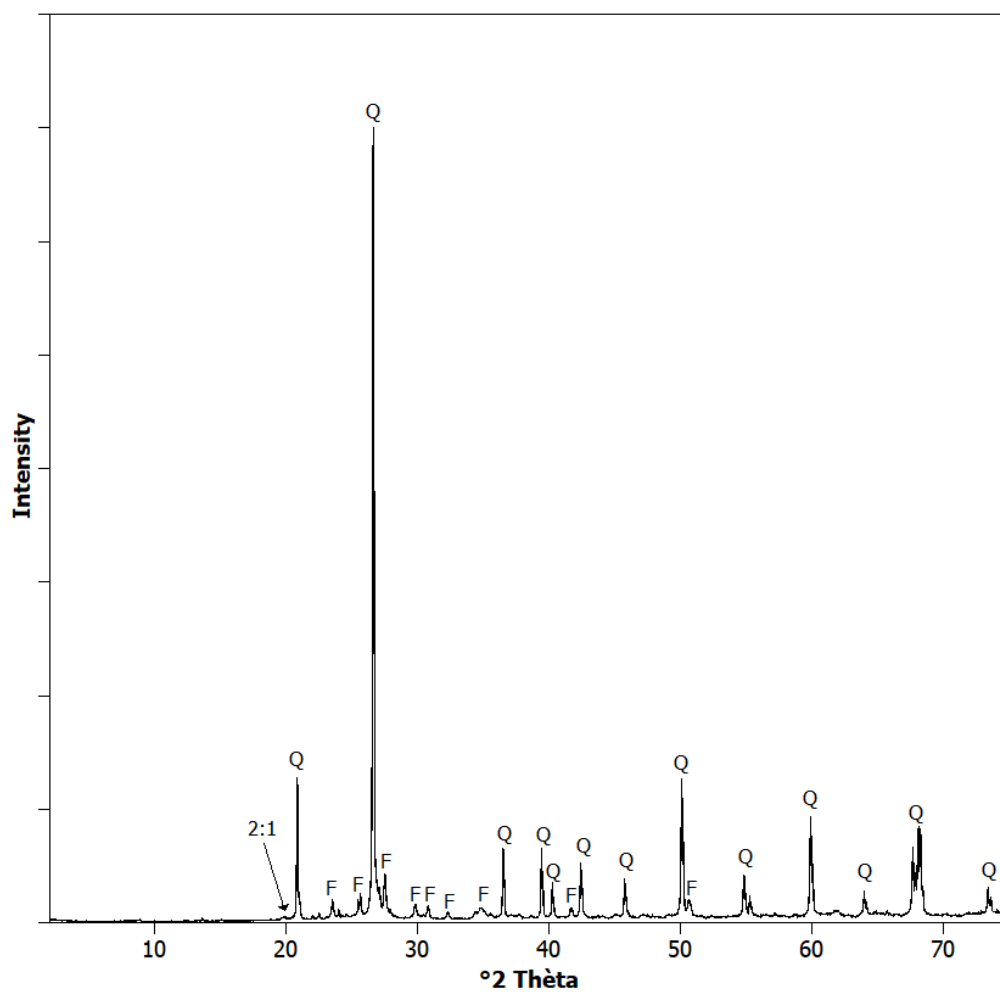


Figure 3: Diffraction pattern of the sample "S Uphill". The minerals that contribute to the main reflections are labeled: 2:1 layer silicates (2:1), Quartz (Q), Alkali feldspar and Plagioclase (F).

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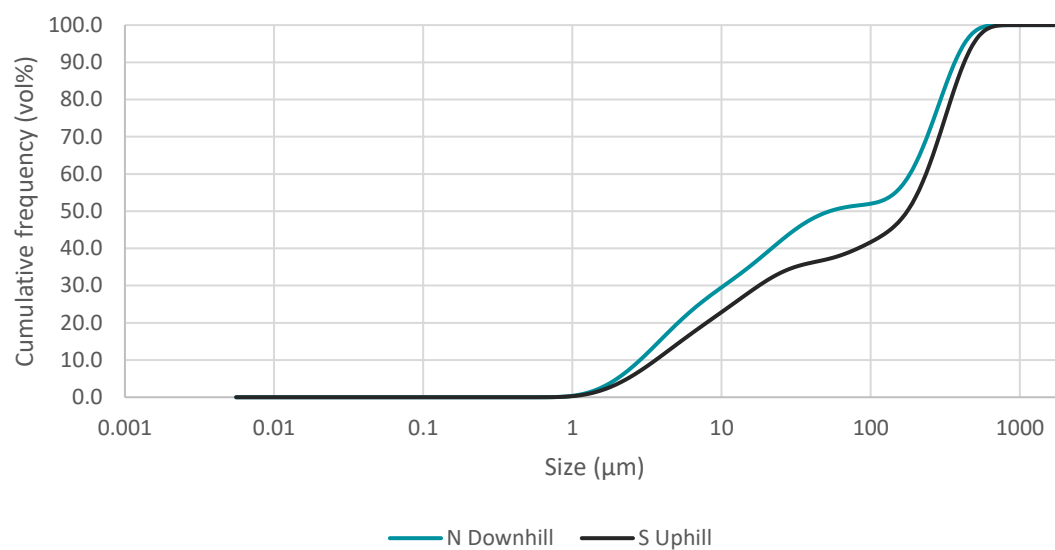


Figure 4. Cumulative frequency distribution of the samples.